



1.40 Water Truck Operation

1. Purpose

Provide basic instructions to Silverpoint employees for loading and unloading a water truck

2. Scope

This procedure is intended for use by all Silverpoint equipment operators

3. Hazard Assessment & Identification

- Personal injury
- Slips, trips, falls
- Physical stress
- Mental stress
- Open pits
- Equipment stress /failure
- Lifting / bending
- Hose movement from changing pump direction
- Hose securement
- Weather extremes
- Steep slopes [which may require a harness and lanyard]
- Thin ice
- Wildlife

4. EHS Requirements

- Fire retardant coveralls
- Steel toed boots
- Safety Glasses
- Hard hat
- Gloves



- Chock blocks
- Hearing Protection (if required)
- Harness and lanyard (if required)
- Life jacket (if required)
- Hose screen (if required)

5. References

- Silverpoint Corporate Codes of Practice
- Alberta OH&S
- Silverpoint Task List
- Silverpoint THA 1.40

6. Procedure

6.1. Engaging PTO

- 6.1.1. Push the clutch to the floor and place the transmission into gear to stop the transmission from turning
- 6.1.2. Engage PTO and take transmission out of gear
- 6.1.3. Slowly release clutch, listen for any type of "grinding noise" from the PTO, if there is no noise bring truck to the proper operating RPM
- 6.1.4. If there is a "grinding noise" push the clutch in and disengage the PTO, repeat the above steps

6.2. Pump operation

- 6.2.1. Use the hydraulic control lever inside the pump box to control loading or unloading truck
- 6.2.2. Slowly push or pull the lever to the desired direction for loading or unloading
- 6.2.3. Always pause in the neutral position for a few seconds to allow the water to stop moving when changing pump direction to prevent the hose from jumping or damaging the pump or hydraulics

Loading from an open {unpressurized } water source [i.e., Dugouts , pits, lakes, streams , rivers .

Always check with the Environmentalist about what size of screen to use when loading from a water source .

All references to load and unload are referring to the truck, i.e., loading the truck or unloading the truck.



- 6.2.4. Park in safe area, use a spotter, if possible, when backing up is required, apply brakes and put chock blocks under tires, engage PTO and raise engine speed to proper rpm
- 6.2.5. Connect hose(s) to truck
- 6.2.6. Place the hose out of the operator's area, best practice is to have the hose in the fender hooks and chained in place to prevent movement
- 6.2.7. Ensure dump valve and passenger side valve (if equipped) are closed
- 6.2.8. Close drain valve on pump
- 6.2.9. Open belly valve (upper left side of pump compartment or air switch in cab)
- 6.2.10. Open valve on hose side of pump
- 6.2.11. Prime pump (add water to pump to create suction and to keep bushings cool)
- 6.2.12. Ensure that you can safely access water source [e.g., steep slopes, thin ice]
- 6.2.13. Attach screen to hose (usually a screen with 3mm holes, sometimes in a dugout you can use a screen with larger holes)
- 6.2.14. Put screen with hose attached into water (source determines how deep to put screen e.g., deep enough to not create a surface vortex but not so deep as to suck mud into screen) It is always a good idea to tie a float (e.g., Empty washer fluid jug) close to the screen to prevent sucking off the bottom of the source. (Mud and debris will damage the pump)
- 6.2.15. Move pump control to load (all the air has to be sucked through the pump before water flows, which means fittings have to seal)
- 6.2.16. Grease pump as required (Every load)
- 6.2.17. Fill to desired level
- 6.2.18. Move pump control lever to neutral, close valve on hose side of pump
- 6.2.19. Open cam lock ear to let air into hose, drain hose and disconnect. In cold weather ensure to remove hose from water source so it does not freeze into the hole.
- 6.2.20. Any hoses left connected to the pump while traveling must be capped
- 6.2.21. Lower rpm of engine and disengage PTO

6.3. Loading from a closed source [pressurized]

- 6.3.1. Loading thru rear vent/load line

- 6.3.1.1. Park in safe area, use a spotter, if possible, when backing up is required, apply brakes and put chock blocks under tires



- 6.3.1.2 . Ensure rear vent/load line valve is open (if equipped)
- 6.3.1.3 . Close drain valve on pump
- 6.3.1.4 . Close belly valve , 4 in valve on passenger side of truck and dump valve (if equipped)
- 6.3.1.5 . Attach hose to rear vent pipe and water source
- 6.3.1.6 . Open-source valve and fill to desired level
- 6.3.1.7 . Close source valve
- 6.3.1.8 . Drain water from line
- 6.3.1.9 . Disconnect hose
- 6.3.2 . Loading thru 4-inch gravity drain (if equipped)
 - 6.3.2.1 . Park in safe area and apply brakes
 - 6.3.2.2 . Close drain valve on pump
 - 6.3.2.3 . Connect hose to 4-inch drain valve and source
 - 6.3.2.4 . Open 4-inch drain valve and belly valve
 - 6.3.2.5 . Open water source valve and fill to desired level
 - 6.3.2.6 . Shut source valve then shut 4-inch gravity valve
 - 6.3.2.7 . Disconnect and drain hose

6.4. Loading from a 400bbl tank

- 6.4.1 . Park in safe area , use a spotter , if possible , when backing up is required , apply brakes and put chock blocks under tires , engage PTO and raise engine speed to proper rpm
- 6.4.2 . Place the hose out of the operator's area , best practice is to have the hose in the fender hooks and chained in place to prevent movement
- 6.4.3 . Close 4-inch valve on passenger side of truck and dump valve (if equipped)
- 6.4.4 . Close drain valve on pump
- 6.4.5 . Attach proper fitting to 400bbl tank (if you do not know if the tank is clean use the highest outlet to prevent solids from going thru the pump)
- 6.4.6 . Attach hose to outlet , attach hose to truck
- 6.4.7 . Open belly valve
- 6.4.8 . Open valve on hose side of pump



- 6.4.9. Open valve on 400bbl tank outlet
- 6.4.10. Move pump control to load
- 6.4.11. Grease pump as required (every load)
- 6.4.12. Fill truck to desired level
- 6.4.13. Close source valve and open camlock to allow air into hose
- 6.4.14. Disconnect from 400 bbl tank and drain fluid from hose towards the pump
- 6.4.15. Close valve on hose side of pump and put pump in neutral
- 6.4.16. Lower engine rpm and disengage PTO

Bungee cords/straps are NOT an acceptable method of securing a hose.

When unloading it is always preferred to connect to a camlock , if a camlock is not available the hose must be secured by tying it to the tank with rope or a proper tie down .

6.5. Unloading in to Rig Tank, flock tank or mud tank

- 6.5.1. Park in safe area, use a spotter, if possible , when backing up is required , apply brakes and put chock blocks under tires, engage PTO and raise engine speed to proper rpm
- 6.5.2. Connect hose to truck
- 6.5.3. Place the hose out of the operator's area, best practice is to have the hose in the fender hooks and chained in place to prevent movement
- 6.5.4. Secure hose to the proper camlock on the tank
- 6.5.5. Open tank valve if there is one
- 6.5.6. Open valve on hose side of pump
- 6.5.7. Open belly valve
- 6.5.8. Move pump control valve to unload (check for leaks)
- 6.5.9. Fill tank to desired level
- 6.5.10. Put control valve in neutral to stop water flow and close tank valve if equipped
- 6.5.11. Move pump control to load, open one camlock ear to allow air into hose, disconnect and drain hose(s) back



to truck

6.5.12. Move pump control to neutral (if there still water on truck close valve on hose side of pump)

6.5.13. Remove hose from tank

6.5.14. Lower engine rpm and disengage PTO

6.6. Unloading into 400 bbl tank

Do not load into overflow line of a 400 bbl tank unless it can be verified that the tank has a top vent, check with site supervisor. Do not climb onto the tank for any reason as you do not have the proper training or PPE

6.6.1. Park in safe area and apply brakes and put chock blocks under tires, engage PTO and raise engine speed to proper rpm

6.6.2. Ensure hose connection to truck is secure

6.6.3. Attach camlock fitting to bottom valve of the 400bbl tank

6.6.4. Attach hose to camlock fitting on tank

6.6.5. Place the hose out of the operator's area, best practice is to have the hose in the fender hooks and chained in place to prevent movement

6.6.6. Open belly valve (upper left side of pump compartment or air switch in cab)

6.6.7. Open valve on 400bbl tank

6.6.8. Open valve on hose side of pump

6.6.9. Move pump control valve to unload (check for leaks)

6.6.10. Fill tank to desired level

6.6.11. When done unloading put the pump in neutral and close truck valve

6.6.12. Close 400bbl tank valve

6.6.13. Re-open truck valve

6.6.14. Move pump control valve to load, slowly undo camlock on 400bbl tank and drain hose back to the truck

6.6.15. Close truck valve and move pump control to neutral

6.6.16. Remove hose from tank

6.6.17. Remove camlock fitting

6.6.18. Lower engine rpm and disengage PTO



It is crucial that the tank is vented at all times, ensure that the vent line is open and the hatch on top of the tank is open and secured. These need to be monitored in the winter for ice buildup and kept clear. If the vents do become plugged the tank can be sucked in when off-loading as no air can get into tank.

Remember : if you are in doubt about what to do, call your Supervisor